RESEARCH-LED EDUCATIONAL TECHNOLOGY / INNOVATION COMPETITION POSTER

HUMAN-CENTRED RESPONSIBLE GENERATIVE AI MICROCREDENTIALS WITH ADAPTIVE LEARNING

CulturaGenAI Academy + m-DOAI AdaptEngine + NADI | CADG-informed | Proposed m-DOAI

v3.2 • 08 NOV 2026

02	06	04	39	18
PUBLIC DEMOS	CORE MODULES	LANGUAGES	SOURCE CHECKS	FINAL EXAM ITEMS

01 ABSTRACT & EDUCATIONAL NEED

A focused learning ecosystem

CulturaGenAI Academy combines six responsible GenAI modules with culturally appropriate design and explainable adaptive practice. The separate AdaptEngine recommends activities, while educators retain authority over learner evidence and completion decisions. [1–3,5]

02 PROBLEM: IDEAL → REALITY → IMPACT → RESPONSE

01 IDEAL — Every learner should receive accessible, culturally relevant GenAI education with ethical guidance and verifiable competence.

02 CURRENT — Learner readiness varies, and AI-generated claims, privacy risks and cultural context require deliberate teaching. [1–3]

03 IF UNRESOLVED — Learners may over-trust outputs or miss biases, while educators lack comparable evidence for targeted support.

04 OUR RESPONSE — The Academy links CADG-informed lessons, adaptive practice, m-DOAI reflection and educator-reviewed evidence.

03 RESEARCH GAP & GUIDING QUESTION

Integration and validation gap

International guidance defines AI competencies and ethical safeguards; knowledge tracing estimates mastery. The combined CADG–m-DOAI microcredential workflow still needs evaluation for cultural suitability, learning outcomes, fairness and transferability. [1–5]

Can culturally responsive, explainable adaptation support responsible GenAI competence?

04 RESEARCH & DEVELOPMENT OBJECTIVES

1. Develop six aligned, assessable responsible GenAI microcredentials.
2. Operationalise CADG and the proposed D–O–A–I constructs in learning.
3. Demonstrate explainable adaptive practice and its decision rationale.
4. Enable multilingual evidence, human review and governed completion.

05 CORE INNOVATION: CADG + m-DOAI + ADAPTENGINE + NADI

CADG
Culture · language
Accessibility · usability
Responsible content

→ **D** DESIGN: Human-centred learning
O OPPORTUNITY: Generative exploration
A ACTIVITY: Iterative human–AI work
I INTELLIGENCE: Human-regulated AI contribution

→ **ADAPT/NADI**
Rules + BKT
Offline NADI demo
Optional secure LLM

CADG guides cultural appropriateness. The proposed m-DOAI framework structures human learning activity; established algorithms suggest next practice, and NADI offers bounded learning guidance.

06 LEARNER JOURNEY: EVIDENCE BEFORE COMPLETION

1	2	3	4	5	6
LEARN	PRACTISE	DOCUMENT	VERIFY	REVIEW	COMPLETE
Six modules	Rules + BKT	D/O/A/I	Source checks	Educator	Pilot record

07 ONE OFFICIAL CURRICULUM: SIX MICROCREDENTIALS

- **M01** GenAI Foundations & Human Agency — Learners explain GenAI limitations and retain responsibility for final decisions.
- **M02** Privacy, Consent & Safe AI Use — Learners protect personal information and practise informed, safe disclosure.
- **M03** Academic Integrity, Disclosure & Verification — Learners verify claims, check references and acknowledge AI assistance.
- **M04** CADG, Bias, Culture & Inclusive AI — Learners evaluate cultural fit, potential bias and accessible design.
- **M05** Responsible Prompting & Human–AI Iteration — Learners improve prompts through comparison, verification and reflection.
- **M06** AI Governance, Assurance & Ethical Leadership — Learners apply human oversight, risk controls and accountable decision-making.

08 METHODOLOGY & CURRENT TECHNICAL VERIFICATION

1	2	3	4	5
ANALYSE	DESIGN	BUILD	CHECK	VALIDATE
Requirements and research guidance	CADG + proposed m-DOAI mapping	MVC demos and secure pilot code	39 source tests + public audit	Future consented learner pilot

Human review, ethics and security acceptance testing are part of the planned validation—not verified learner outcomes.

09 DEMONSTRATED SOFTWARE & RESULTS



CulturaGenAI Academy
Six-module learning and D/O/A/I evidence



m-DOAI AdaptEngine
Independent adaptive-practice component

Source verified: 39 automated checks passed. Outputs use synthetic learner data.

10 SECURITY-BY-DESIGN: PILOT CODE, NOT CERTIFICATION

- ✓ **AUTH** — Supabase authentication and institution membership roles.
- ✓ **ISOLATION** — RLS policies and server-side grading in private backend code.
- ✓ **OVERSIGHT** — Educator evidence approvals and auditable completion records.
- ! **BOUNDARIES** — NADI cannot issue credentials or change grades.

11 CONTRIBUTIONS TO THEORY & PRACTICE

THEORY — A testable operational linkage of CADG, the proposed m-DOAI constructs and explainable adaptation; no new theory is claimed as validated.

PRACTICE — Reusable learning code, four-language access, source-verification tasks, bounded NADI guidance and an educator-controlled completion workflow.

12 LIMITATIONS, PILOT PLAN & SCALABILITY

CURRENT LIMITATIONS

BKT probabilities remain provisional. No live Supabase acceptance test, independent security audit, accredited credential, learning-impact evaluation or TRL 9 evidence has been completed.

NEXT VALIDATION

Recruit a consented pilot; review item content, language and CADG fit; test usability, learning gains, fairness, security, staff workload and operating cost.

SELECTED REFERENCES & RESEARCH LINEAGE

[1] Miao & Holmes (2023), UNESCO GenAI guidance. [2] Miao, Shiohira & Lao (2024), UNESCO AI competencies for students.
[3] Miao & Cukurova (2024), UNESCO AI competencies for teachers.
[4] Corbett & Anderson (1994), Knowledge tracing, DOI: 10.1007/BF01099821. [5] ISO 9241-210:2019, Human-centred design.
Research lineage: CADG instrument and proposed m-DOAI technical report (internal research; extension requires validation).